FIITJEE

ALL INDIA TEST SERIES

FULL TEST – II

JEE (Main)-2020

TEST DATE: 22-12-2019

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. During a rainy day, rain is falling vertically down with a velocity 2 m/s. A boy at rest starts his motion with a constant acceleration of 2 m/s^2 along a straight road. Find the rate at which the angle of the axis of the umbrella with vertical should be changed so that the rain always appears to fall parallel to the axis of the umbrella.

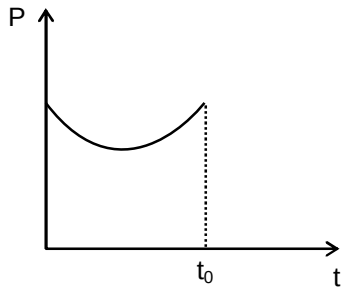
(A) $\frac{1}{1+t^2}$

(B) $\frac{2}{1+t^2}$

(C) $\frac{1}{2+t^2}$

(D) $\frac{1}{1+2t^2}$

2. Power versus time graph for a given force is shown in the figure. Then work done by the force upto time $t(\le t_0)$.

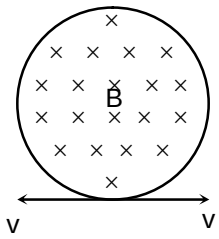


(A) First increases then decreases.
(B) First decreases then increases.
(C) Always increases.
(D) Always decreases.

3. A cobalt (atomic no. = 27) target is bombarded with electrons and the wave lengths of its characteristic X-rays spectrum are measured. A second weak characteristic spectrum is also found due to an impurity in the target. The wavelength of the K_α lines are 225.0 pm (cobalt) and 100.0 pm (impurity). Atomic number of the impurity is

(A) 39
(B) 40
(C) 59
(D) 60

4. A circular conducting loop of radius r_0 and having resistance λ per unit length as shown in the figure is placed in a magnetic field B which is constant. The ends of the loop are crossed and pulled in opposite directions with a velocity v such that the loop always remains circular and the radius of the loop goes on decreasing, then



(A) current induced in the loop is $\frac{Bv}{2\pi\lambda}$.

(B) current induced in the loop is $\frac{Bv}{\pi\lambda}$.

(C) current induced in the loop is $\frac{2Bv}{\pi\lambda}$.

(D) current induced in the loop is $\frac{Bv}{2\lambda}$.

5. When a particle of mass m is suspended vertically by a spring of unstretched length 'a', the spring is stretched to a total length 'ℓ'. The particle is pulled down by a small distance 'b' from equilibrium position and then released. The angular frequency of the simple harmonic motion executed by the particle is

(A) $\sqrt{\frac{g\ell}{(\ell-a)^2}}$

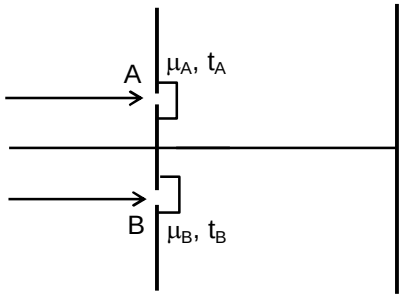
(B) $\sqrt{\frac{ga}{(\ell-a)^2}}$

(C) $\sqrt{\frac{g}{(\ell-a)}}$

(D) $\sqrt{\frac{gb}{(\ell-a)^2}}$

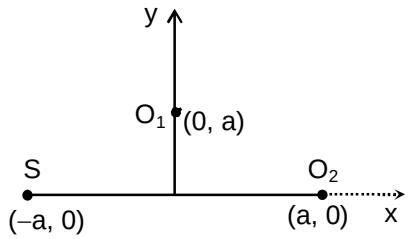
6. In YDSE, let A and B be two slits as shown. Thin films of thickness t_A and t_B and refractive indices μ_A and μ_B are placed in front of A and B respectively. If $\mu_A t_A = \mu_B t_B$, then the central maxima will

(A) not shift
(B) shift towards A, if $t_B > t_A$
(C) shift towards B, if $t_B > t_A$
(D) shift towards A, if $t_A > t_B$



7. A sound source S and observer O_1 and O_2 are placed as shown in figure. S is always at rest and O_1 and O_2 start moving along positive y-axis and positive x-axis respectively with the same constant speed v_0 at t = 0 away from S. At t = 5 sec, f_1 and f_2 represent apparent frequencies of sound received by O_1 and O_2 respectively. Then the ratio $\left(\frac{f_1}{f_2}\right)$ is

(A) zero
(B) between zero and one
(C) one
(D) more than one

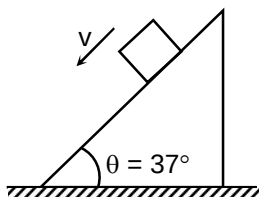


8. A cubical block of side 'a' and density 'ρ' slides down a fixed inclined plane of inclination $\theta = 37°$ with a constant velocity v as shown. There is a thin film of viscous liquid of thickness 't' between the inclined plane and the block. Then the coefficient of viscosity of the liquid will be

(A) $\frac{3\rho agt}{5v}$

(B) $\frac{4\rho agt}{5v}$

(C) $\frac{\rho agt}{v}$

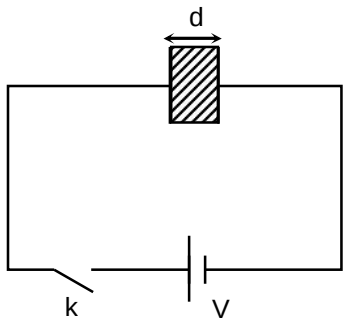


(D) $\frac{5\rho agt}{3v}$

9. A diverging lens of focal length f_1 is placed infront of and coaxially with a concave mirror of focal length f_2. The separation between the lens and the mirror is d. A parallel beam of light incident on the lens returns as a parallel beam from the arrangement.
(A) The beam diameter of the incident and reflected beams will not be same.
(B) $d = 2|f_2| - |f_1|$
(C) If the entire arrangement is immersed in water, the conditions will remains unaltered.
(D) none of the above

10. A parallel plate capacitor with initial plate distance d_0 is charged upto voltage V_0 and then detached from the charger. Now the distance between the plates is varied periodically according to the law $d = d_0 + d_1 \sin \omega t$. The value of d_1 if the potential difference across the capacitor changes according to the law
$V = V_0\left(1+\frac{1}{2}\sin\omega t\right)$ is
(A) d_0
(B) $\frac{d_0}{2}$
(C) $\frac{3d_0}{2}$
(D) $2d_0$

11. The area of each plate of a capacitor is S and it is completely filled with a dielectric as shown in figure. The dielectric constant and resistivity of the material of the slab inserted in the capacitor are K and ρ respectively. The emf of the ideal cell is V. The current flowing through dielectric slab immediately after the key is switched on is



(A) $\frac{V}{\rho d}S$
(B) $\frac{2V}{\rho d}S$
(C) $\frac{V}{2\rho d}S$
(D) zero

12. A thin insulating rod of length ℓ carries a charge q distributed uniformly on it. The rod is pivoted at its mid point and is rotated at a frequency f about a fixed axis perpendicular to the rod and passing through the pivot. The magnetic moment of the rod system is
(A) $\frac{1}{12}\pi qf\ell^2$
(B) $\pi qf\ell^2$
(C) $\frac{1}{6}\pi qf\ell^2$
(D) $\frac{1}{3}\pi qf\ell^2$

13. A particle of charge q and mass m starts moving from the origin under the action of an electric field $\vec{E} = E_0\hat{i}$ and $\vec{B} = B_0\hat{i}$ with a velocity $\vec{v} = v_0\hat{j}$. The speed of the particle will become $2v_0$ after a time

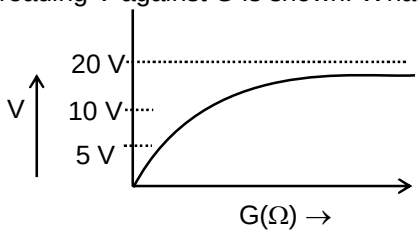
(A) $t = \dfrac{2mv_0}{qE_0}$

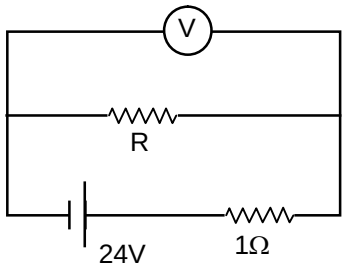
(B) $t = \dfrac{2B_0q}{mv_0}$

(C) $t = \dfrac{\sqrt{3}B_0q}{mv_0}$

(D) $t = \dfrac{\sqrt{3}mv_0}{qE_0}$

14. A cell of internal resistance 1 Ω is connected across a resistor R. A voltmeter having variable resistance G is used to measure potential difference across the resistor. The plot of voltmeter reading V against G is shown. What is the value of external resistor R?



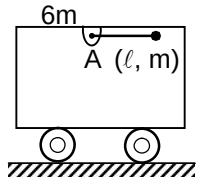


(A) 5 Ω
(B) 4 Ω
(C) 3 Ω
(D) 1 Ω

15. Consider the equation $\dfrac{d}{dt}\left[\int \vec{F}\cdot d\vec{s}\right] = A\left[\vec{F}\cdot\vec{P}\right]$. Then dimensions of A will be (where $\vec{F}$ = force, $d\vec{s}$ = small displacement, t = time and $\vec{P}$ = linear momentum)

(A) $[M^0L^0T^0]$
(B) $[M^1L^0T^0]$
(C) $[M^{-1}L^0T^0]$
(D) $[M^0L^0T^{-1}]$

16. In the figure shown the cart of mass 6 m is initially at rest on a smooth horizontal surface. A particle of mass m is attached to the end of the light rigid rod which can rotate freely about end A in a vertical plane. If the rod is released from rest from the horizontal position as shown, find the velocity (v_{rel}) of the particle with respect to the cart when the rod is vertical.



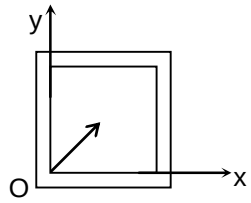
(A) $\sqrt{\dfrac{7}{3}g\ell}$

(B) $\sqrt{\dfrac{7}{6}g\ell}$

(C) $\sqrt{\dfrac{14}{3}g\ell}$

(D) $\sqrt{\frac{8}{3}g\ell}$

17. The figure shows a square carrom board without any pockets. A coin is pushed from the corner, which is the origin, with a velocity $\vec{v} = \left(2\hat{i} + 3\hat{j}\right)$. Assume gravity and friction to be absent. The coin collides with edges of the carrom board elastically. What is the velocity vector of coin after the 3rd collision?



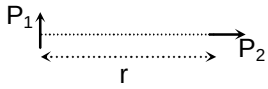
(A) $2\hat{i} + 3\hat{j}$

(B) $2\hat{i} - 3\hat{j}$

(C) $-2\hat{i} + 3\hat{j}$

(D) $-2\hat{i} - 3\hat{j}$

18. Two point electric dipoles with dipole moments P_1 and P_2 are separated by a large distance 'r' with their dipole axes mutually perpendicular as shown. The electric force between the dipoles is (where $K = \frac{1}{4\pi\varepsilon_0}$)



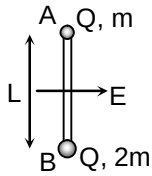
(A) $\frac{6KP_1P_2}{r^4}$

(B) $\frac{3KP_1P_2}{r^4}$

(C) $\frac{2KP_1P_2}{r^4}$

(D) $\frac{KP_1P_2}{r^4}$

19. Two small balls A and B of positive charge Q each and masses m and 2m respectively are connected by a non conducting light rigid rod of length L. This system is released in a uniform electric field of strength E as shown. Just after the release (assume no other force acts on the system).



(A) rod has zero angular acceleration.

(B) rod has angular acceleration $\frac{QE}{2mL}$ in anticlockwise direction.

(C) acceleration of point A is $\frac{2QE}{m}$ towards right

(D) acceleration of point A is $\frac{QE}{m}$ towards right

20. The plane polarised light is passed through a polaroid. On viewing through the polaroid, we find that when the polaroid is given a complete rotation about an axis coinciding with the direction of the incident light, one of the following is observed,

(A) the intensity of light gradually decreases to zero and remains zero.

(B) the intensity of light gradually increases to a maximum and remains maximum.

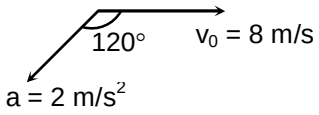
(C) there is no change in intensity

(D) the intensity of light is twice maximum and twice zero.

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. The following figure shows the velocity and acceleration of a point like body at the initial moment of its motion. The acceleration vector of the body remains constant. The time (in sec) after which its speed becomes 8 m/s again is

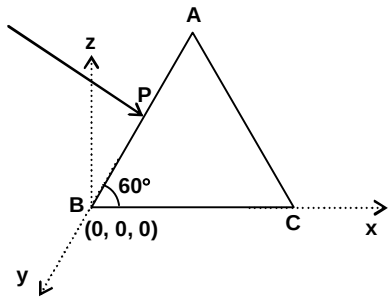


22. A rod has variable coefficient of linear expansion $\alpha = \frac{x}{5000}$, where x is the distance from one end of the rod. If the length of the rod is 1m then find the increase in length of the rod (in cm) on increasing temperature of the rod by 100°C.

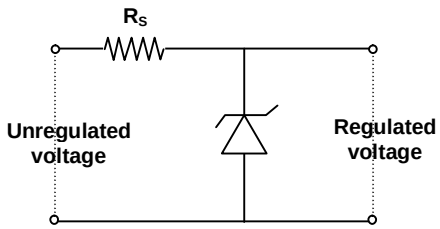
SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. An equilateral prism ABC having edge length 10 cm is placed in air with its base BC lying horizontally along x-axis as shown in figure. A ray of light represented by equation $\sqrt{3}z + x = 10$ is incident at a point P on the face AB of the prism. Find the value of μ for which the ray grazes the face AC.



24. A Zener of power rating 1 W is to be used as voltage regulator. If Zener has a breakdown of 5V and it has to regulate voltage which fluctuated between 3V and 7V, what should be the value of R_S(in Ω) for safe operation as shown in the figure.



25. A stone is projected at an angle 45° with horizontal. A bee follows the trajectory of the stone at a constant speed equal to the initial speed of the stone. Neglect air resistance for motion of stone. Find the magnitude of acceleration (in m/s^2) of the bee at the topmost point of its trajectory: (g = 9.8 m/s^2)

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. Which of the following relation is correct regarding one mole of an ideal gas for adiabatic reversible process?

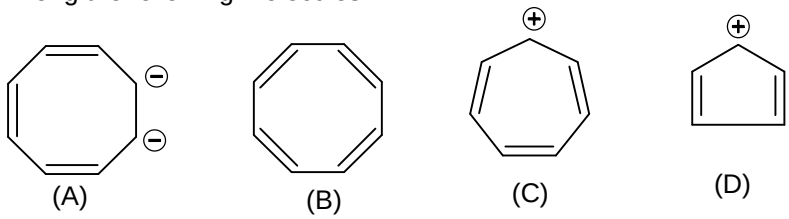
(A) $\ln\frac{T_2}{T_1}=\frac{R}{C_V}\ln\frac{V_2}{V_1}$

(B) $\ln\frac{T_2}{T_1}=\frac{R}{C_P}\ln\frac{P_1}{P_2}$

(C) $\ln\frac{T_2}{T_1}=\frac{R}{C_P}\ln\frac{P_2}{P_1}$

(D) $\frac{R}{C_V}\ln\frac{V_1}{V_2}=\frac{R}{C_P}\ln\frac{P_1}{P_2}$

27. Among the following molecules



(A) (B) (C) (D)

Choose the correct option:

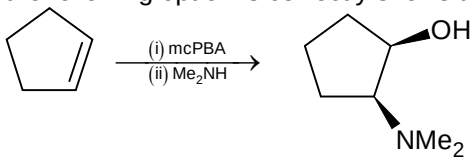
(A) 'A' and 'B' are non-aromatic.
(B) 'B' and 'D' are antiaromatic.
(C) 'B' is antiaromatic and 'C' is aromatic.
(D) 'A' is an aromatic and 'B' is non-aromatic.

28. Which of the following statement is not correct?

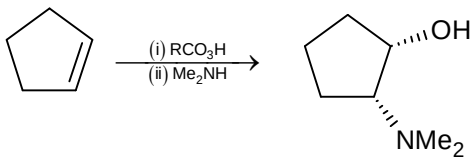
(A) Except glycine all other α-amino acids are optically active.
(B) At isoelectric point solubility of amino acid in H_2O is minimum
(C) At isoelectric point molecule has no net change.
(D) All disaccharides are reducing sugar.

29. Which of the following option is correctly shows the major product of given reaction?

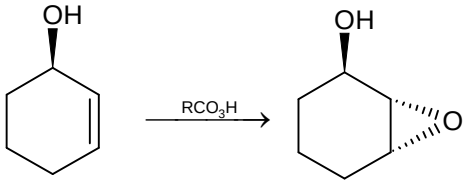
(A)



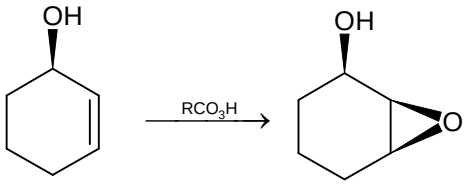
(B)



(C)



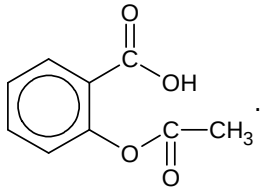
(D)



30. An acid base indicator (HIn) is 50% ionized at pH = 6.0. What will be the dissociation constant of HIn?
(A) 10^{-6}
(B) 10^{-5}
(C) 10^{-4}
(D) 0.5×10^{-6}

31. Which of the following order is incorrect?
(A) $F_2 < Cl_2 < Br_2 < I_2$ [Bond length]
(B) $F_2 > Cl_2 > Br_2 > I_2$ [Oxidizing power]
(C) $F_2 > Cl_2 > Br_2 > I_2$ [Bond dissociation energy]
(D) $F_2 < Cl_2 < Br_2 < I_2$ [Boiling point]

32. A crystalline solid has three types of atoms X, Y and Z. X forms fcc lattice, Y at each octahedral voids and Z at each tetrahedral voids. If all the atoms along one body diagonal and from one face has been removed. What will be the formulae of compound?
(A) $X_{22}Y_{16}Z_{48}$
(B) $X_{23}Y_{16}Z_{48}$
(C) $X_{11}Y_4Z_{24}$
(D) $X_{15}Y_6Z_8$

33. Which of the following statement is wrong about aspirin?
(A) It has anti blood clotting action.
(B) Its structure is 

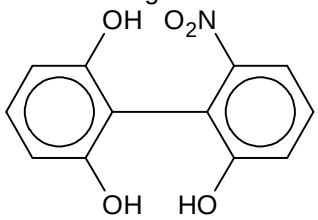
.
(C) It is a pain killer.
(D) It is a narcotic drug.

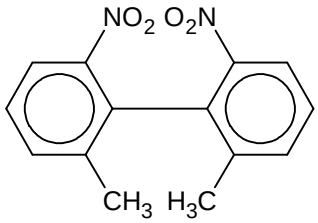
34. According to law of equipartition of energy. How many degree of freedoms are active for energy distribution in SO_2 (g) molecule at low temperature.
(A) 6
(B) 5
(C) 9
(D) 3

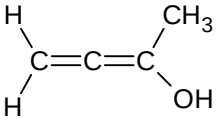
35. Among NF_3 and NCl_3 which statement is not correct?
(A) Melting point of NF_3 is lower than NCl_3.
(B) Bond angle of F – N – F is greater than Cl – N – Cl.

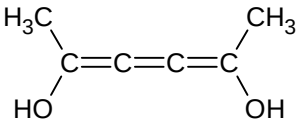
(C) The bond length of N – F is shorter than N – Cl.
(D) Dipole moment of NCl_3 is greater than NF_3.

36. Choose the correct statement.
(A) If a gas gets adsorbed on the surface of a solid, the gas and the solid are known as adsorbent and adsorbate respectively.
(B) The entropy of adsorbed gas molecule is higher than that of the molecule in gaseous state.
(C) Solutes causing positive adsorption are called surface active agent.
(D) Chemiosoption is independent of temperature.

37. Which of the following molecule is optically active?
(A) 

(B) 

(C) 

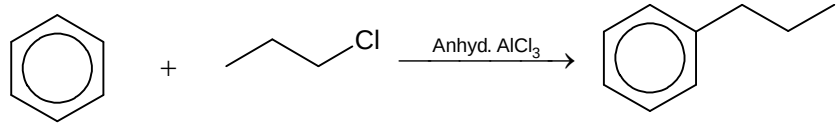
(D) 

38. 5 litre of an unknown ideal gas weigh 10 g at 1.5 atm pressure. What will be the RMS speed of unknown gas molecule.
(A) 477 m/s
(B) 492 m/s
(C) 323 m/s
(D) 172 m/s

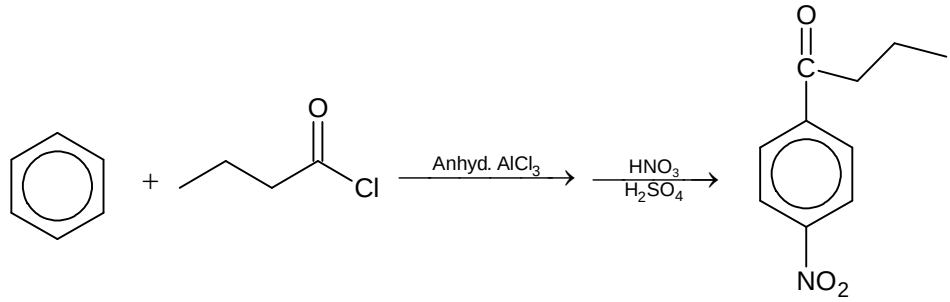
39. Relative reactivity of carbonyl compounds towards nucleophile?
(A) Acyl halide > Acid anhydride > Aldehyde > Carboxylic acid.
(B) Carboxylic acid > Aldehyde > Acid anhydride > Acyl halide.
(C) Acyl halide > Aldehyde > Acid anhydride > Carboxylic acid.
(D) Acid anhydride > Acyl halide > Carboxylic acid > Aldehyde.

40. Which of the following reaction is correct for major product?

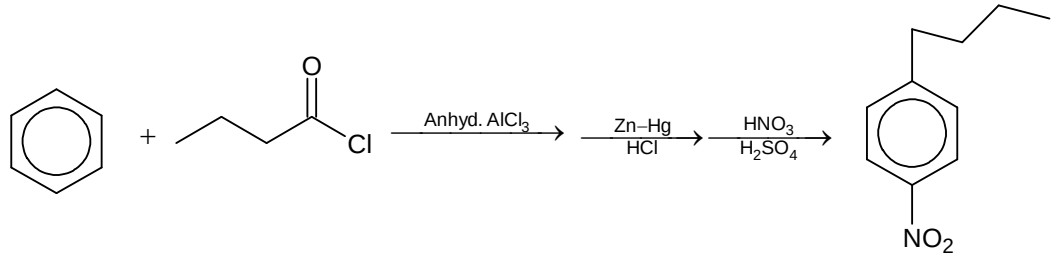
(A)



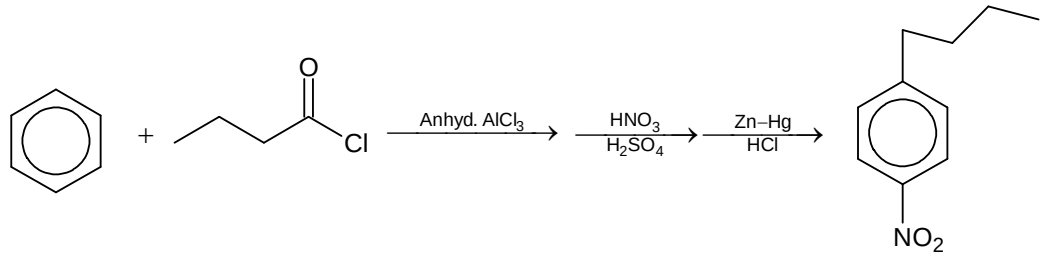
(B)



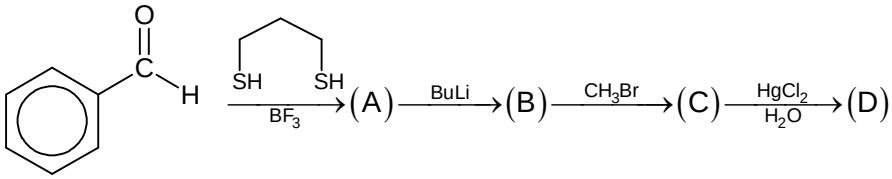
(C)



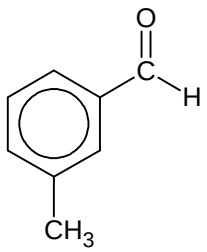
(D)

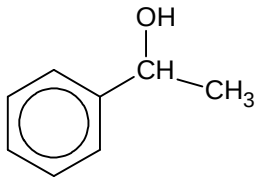


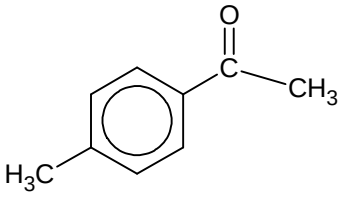
41. The major product 'D' will be:

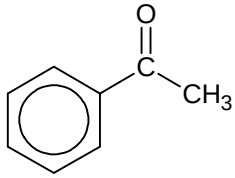


(A)

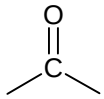


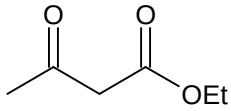
(B) 

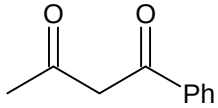
(C) 

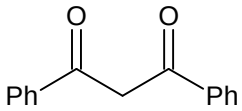
(D) 

42. Which of the following molecule has maximum enol content in their pure liquid state?

(A) 

(B) 

(C) 

(D) 

43. Which of the following statemet is false?

(A) In photoelectric effect, the number of electrons ejected is proportional to the intensity of the light used.

(B) In any orbital, there is 100% probability of finding of electron.

(C) The photoelectric effect can't be explained by wave nature of light, but by its particle nature.

(D) Bohr's theory is only applicable for one electron system.

44. The species that does not contains peroxide linkage is

(A) CrO_5

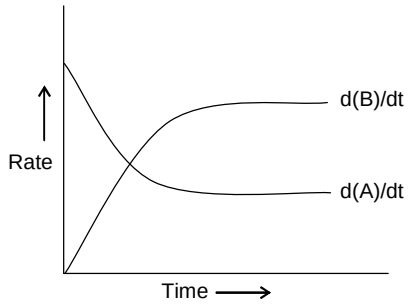
(B) $H_2S_2O_8$

(C) BaO_2

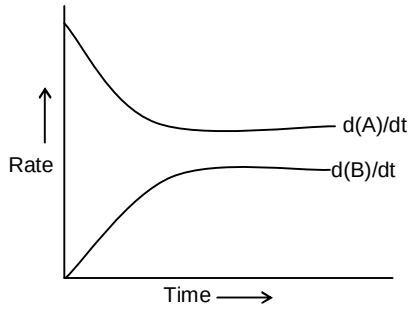
(D) PbO_2

45. Which of the following graph will show the equilibrium condition for the reaction $A \rightleftharpoons B$?

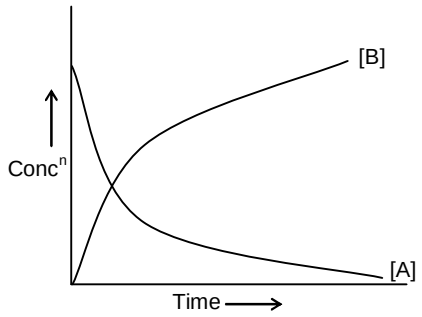
(A)



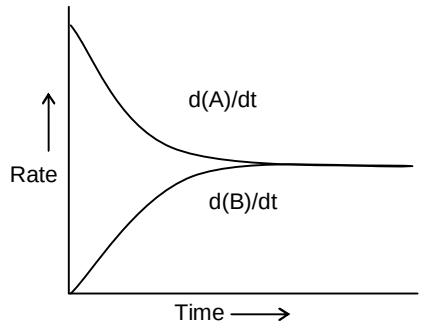
(B)



(C)



(D)

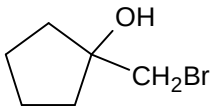


SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. According to crystal field theory, total number of unpaired electrons present in the following molecules

$[NiCl_4]^{2-}$, $[Ni(CN)_4]^{2-}$ and $[Ni(H_2O)_6]^{2+}$ will be

47. 

$\xrightarrow{AgNO_3}(A)\xrightarrow[(ii)\,\Delta]{(i)\text{ dil. NaOH}}(B)$

The degree of unsaturation in major product (B) will be:

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. The equilibrium constant of following reaction at 500 K is 10 atm^{-1}.

$2SO_2(g)+O_2(g)\rightleftharpoons 2SO_3(g)$

What would be the partial pressure of O_2 (in atm) at equilibrium, if at equilibrium the partial pressure of SO_2 (g) is half of that of SO_3 (g).

49. What will be the value of Van't Hoff factor (i) for $K_3\left[Fe(CN)_6\right]$, if it undergoes 75% dissociation in aqueous medium?

50. How many gram of NH_3(g) will be required to add in 500 ml, 0.6 N H_2SO_4 to reduce the normality of H_2SO_4 to 0.1 N.

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. The director circle of a hyperbola is $x^2 + y^2 - 6x - 8y + 5 = 0$. If one end of the transverse axis of hyperbola is (3, –1), then one of the focus of this hyperbola is
(A) $\left(3+5\sqrt{6}, 4\right)$
(B) $\left(3+5\sqrt{30}, 4\right)$
(C) $\left(3,\ 4+\sqrt{30}\right)$
(D) none of these

52. Let α, β, γ be the roots of the equation $10x^3 - mx^2 - 54x - 27 = 0$. If α, β, γ are in H.P., then the value of m is
(A) 6
(B) 7
(C) 9
(D) 18

53. The remainder when $(2017)^{2018} + (2018)^{2019} + (2019)^{2020}$ is divided by 5 is
(A) 2
(B) 4
(C) 3
(D) 0

54. The sum of the series $8! + \frac{9!}{2!} + \frac{10!}{3!} + \frac{11!}{4!} + \ldots..$ upto 20 terms is
(A) $7!\left({}^{27}C_{19} - 1\right)$
(B) $8!\left({}^{27}C_{20} - 1\right)$
(C) $7!\left({}^{28}C_{20} - 1\right)$
(D) $8!\left({}^{28}C_{20} - 1\right)$

55. If A B and C are three non singular square matrices of same order such that $B = CAC^{-1}$, then CA^5C^{-1} is equal to
(A) B
(B) B^{-5}
(C) B^5A^5
(D) B^5

56. Let α_1, α_2 and β_1, β_2 are roots of the equation $ax^2 + bx + c = 0$ and $px^2 + qx + r = 0$ respectively. If the system of equations $\alpha, y + \alpha_2 z = 0$ and $\beta, y + \beta_2 z = 0$ has a non trivial solution, then
(A) $bpr^2 = qac^2$
(B) $b^2pr = q^2ac$
(C) $bp^2r = qa^2c$
(D) none of these

57. The value of $\left|3+\frac{9}{z}\right|^2+|3-z|^2$ if $|z| = 3$ is

(A) 18
(B) 1
(C) 3
(D) 36

58. Tangents are drawn from a point P to the parabola $y^2 = 8x$. If the angle between the tangents is 45°, then point P moves on a curve whose centre is

(A) (12, 0)
(B) (–6, 0)
(C) (–12, 0)
(D) (–8, 0)

59. If $x\,dy - y\,dx = \sqrt{x^2-y^2}\,dx$ is the differential equation of the equation $y = f(x)$. If $f(1) = 0$, then $f\left(e^{\frac{\pi}{2}}\right)$ is

(A) $\frac{\pi}{2}$
(B) 1
(C) $e^{\frac{\pi}{2}}$
(D) 0

60. If a pair of variable straight lines $x^2 + 4y^2 + \alpha xy = 0$ (where α is a real parameter) cut the ellipse $x^2 + 4y^2 = 4$ at two points, then locus of the point of intersection of tangents at A and B is

(A) $4x^2 - y^2 = 0$
(B) $x^2 - 4y^2 = 0$
(C) $x^2 + 4y^2 - 16 = 0$
(D) $x^2 - 4y^2 - 32 = 0$

61. A circle S passes through (0, 1) and is orthogonal to the circles $(x - 1)^2 + y^2 = 16$ and $x^2 + y^2 = 1$. If the radius of S is r and centre is (a, b), then $a + b + r$ is equal to

(A) 0
(B) 1
(C) –7
(D) 9

62. The value of k for which the common tangent in 1st quadrant to the circle $x^2 + y^2 = 16$ and the ellipse $\frac{x^2}{25}+\frac{y^2}{4}=1$ makes an intercept of $\frac{14}{\sqrt{k}}$ between the coordinate axes is

(A) 5
(B) 3
(C) 11
(D) none of these

63. Bag A contains 5 white and 5 black balls. Bag B is empty initially, 5 balls are drawn from bag A and placed in bag B. Then 4 balls are drawn from bag B and placed in an empty bag C. The probability that bag C contains exactly 3 white balls is $\frac{k}{{}^{10}C_5}$, then k is

(A) 20
(B) 80
(C) 60
(D) 33

64. If a, b, c represents the sides opposite to angle A, B, C respectively in ΔABC.
If cos A + 2 cos B + cos C = 2, then a, b, c are in
(A) A.P.
(B) H.P.
(C) G.P.
(D) none of these

65. If the standard deviation of 0, 1, 2, 3, ….. 9 is k, then the standard deviation of 10, 11, 12, 13, ….. 19 is
(A) k
(B) k + 10
(C) $k+\sqrt{10}$
(D) 10 k

66. The inverse of the proposition $(p \wedge \sim q) \rightarrow r$ is
(A) $\sim r \rightarrow (\sim p \vee q)$
(B) $(\sim p \vee q) \rightarrow \sim r$
(C) $r \rightarrow p \wedge \sim q$
(D) $(p \wedge \sim q) \rightarrow \sim r$

67. Let S be the set of real values of λ for which the function $f(x) = x^3 - 3(2\lambda - 1)x^2 + 6\lambda x$ has exactly one local maximum and exactly one local minimum, then S can be
(A) (0, 6)
(B) (1, 4)
(C) $(-\infty, 0)$
(D) $(0, \infty)$

68. The line $\frac{x-1}{\lambda}=\frac{y}{2}=\frac{z+3}{-12}$ makes an isosceles triangle with the planes x + 2y – 3z + 5 = 0 and 2x + 3y – z + 2 = 0, then value of λ can be
(A) 5
(B) 11
(C) 22
(D) 9

69. Let p(x) be a polynomial of degree 3 such that $p(x)=\frac{2}{3+x}$ for x = 0, 1, 2, 3, then p(6) is equal to
(A) 2
(B) 0
(C) –2
(D) –3

70. Let f(x) be a function defined on R such that $f'(x) = 2f'(6 - 2x)$ with f(2) = 4, then f(1) + f(4) is equal to
(A) 4
(B) 0
(C) 8
(D) –4

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. $\lim\limits_{n\to\infty} \dfrac{n\times 5^n}{n(x-3)^n + n\times 5^{n+2} - 5^n} = \dfrac{1}{25}$, ($n \in N$), then the highest positive integral value of x is

72. If α and β are roots of the equation $x^2 - 10x - 6 = 0$ and $P_n = \alpha^n - \beta^n$, then the value of $\dfrac{P_{16} - 6P_{14}}{2P_{15}}$ is equal to

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. $\int \dfrac{(x^4 - x)^{\frac{1}{4}}}{x^5}\,dx = \dfrac{m}{k}\left(\dfrac{x^3 - 1}{x^3}\right)^{\frac{5}{4}} + c$, then k – 2m is equal to (where m and k are coprime)

74. The locus of the image of the focus of the ellipse $9x^2 + 25y^2 = 225$ with respect to any tangent to the ellipse is a circle whose radius is k, then $\dfrac{k}{4}$ is equal to

75. If the portion of the line x + 4y – 3 intercepted between the lines ax – y + 3 = 0 and x + by – 1 = 0 subtends a right angle at the origin, then $\dfrac{2a + 9b}{20}$ is equal to